

JEE MAIN REVISION MASTER: DIFFERENTIATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 83 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Differentiation - Core Application	83 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Differentiation - Core Application (Very High Yield — 83 PYQs)

Core Principle: Key problem pattern in Differentiation.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "If $y = \tan^{-1}\left(\frac{3\cos x - 4\sin x}{4\cos x + 3\sin x}\right) + 2\tan^{-1}\left(\frac{1 + \sqrt{1-x^2}}{x}\right)$, then (dy/dx) at $x = \frac{\sqrt{3}}{2}$ is e..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let $f(x) = x^3 + x^{2f'(1)} + 2xf''(2) + f'''(3)$, $x \in \mathbb{R}$. Then the value of $f'(5)$ is :..." [Ans: N/A]
- [2025 Evening Shift] Q#1 (MCQ): "Let $f: (0, \infty) \rightarrow \mathbb{R}$ be a function which is differentiable at all points of its domain and satisfies the condition $x^2 f'(x) = x$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 2nd April Morning Shift (Q#1)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable odd function satisfying $f'(x) \geq 0, f'''(x) = f(x), f(0) = 0, f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____.

Correct Answer: 36

Solution Analysis: $f'(x) \geq 0, f'''(x) = f(x)$ Second order differential equation $f(x) = Ae^x + Be^{-x}$ $f(0) = 0 \Rightarrow A = -B \Rightarrow f(x) = A(e^x - e^{-x})$ $f'(x) = Ae^x + Ae^{-x} = A(e^x + e^{-x})$ $f'(0) = 3 = A(e^0 + e^0) = 2A \Rightarrow A = (3/2)$ $f(x) = (3/2)(e^x - e^{-x})$ If $\ln 3 = (27/2)(e^{\ln 3} - e^{-\ln 3}) = (27/2)(3 - (1/3)) = (27/2) \cdot (8/3) = 36$